

FOURIER–MUKAI EXCLUSION OF EXPONENTIAL CHARACTERS AT QUARTIC CM FIELDS

ABSTRACT. We prove that on an abelian eightfold of Weil type for a quartic CM field, no perfect complex whose Chern character is a Weil class plus an exponential in the polarisation, with inverse exponent in an explicit lattice, meets the corrected propagation criterion.

1. WHAT THIS NOTE PROVES

The manuscript [BMB] reduces the propagation statement for the split families of a quartic CM field to one object: a perfect complex E on a member A with $\mathrm{ch}(E) = N\omega + P$ and $\dim \mathrm{Ext}^2(E, E) = r(\mathrm{ch} E)$, where ω is a Weil class, P a polynomial in the classes η_s , and r the rank of a contraction ([BMB, Theorem 19.21(v)]; the notation is recalled in §2). The least value a complex meeting this can have is 88 [BMB, Corollary 19.23], and after the Weil tori [BMB, Proposition 19.24, Corollary 19.26] two shapes remain at 88:

$$\begin{aligned} (\mathrm{Exp}) \quad & \mathrm{ch}(E) = N\omega + c + \mathrm{Tr}(a(e^{\lambda\Theta} - 1)), \quad a, \lambda \in F_0^\times, \\ (\mathrm{Lin}) \quad & \mathrm{ch}(E) = N\omega + c + \eta_s, \quad s \in F_0^\times. \end{aligned}$$

The manuscript notes that an autoequivalence of $D^b(A)$ changes the character. This note builds the autoequivalences that act on the two real places separately and uses them.

- (1) *A place-wise Mukai group* (§3). Tensoring by line bundles L_b with $c_1 = \eta_b$, and the conjugates by the Fourier–Mukai transform of tensoring by line bundles M_β on $\hat{A}$, act on $H^*(A)$ through two commuting copies of $\mathfrak{sl}_2$, one at each real place of F_0 , with parameters in two explicit lattices $\mathcal{O}'$ and $\mathfrak{a}$ of F_0 . The transforms by the M_β fix every Weil class, and all of them preserve $\dim \mathrm{Ext}^2$ and r .
- (2) *Exclusion* (Theorem 4.3). If $\lambda^{-1} \in \mathfrak{a}$, equivalently if the inverse of the $\mathbb{Q}$ -homomorphism φ_{η_λ} is a homomorphism $\hat{A} \rightarrow A$, the shape (Exp) is carried to a character that stays of Hodge type on every Weil torus, and the argument of [BMB, Theorem 19.12] excludes it. The same holds at rank 104 with the mixed exponential e^{η_λ} added. The exponential shapes fall into classes indexed by λ^{-1} modulo $\mathfrak{a}$, and the theorem removes the class 0.
- (3) *The linear shape* (§5). No element of $\mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})$ carries (Lin) to a Weil tori character, because it corresponds to x^3y and not to a sum of two fourth powers. Its rank can be shifted by $4 \mathrm{Tr}(s\mathfrak{a})$; its transform by the Weyl element is never a shifted sheaf; on a product member it needs at least four exterior products; and a direct sum reaching 88 has one summand carrying the whole character.

What is not proved: the shape (Exp) with $\lambda^{-1} \notin \mathfrak{a}$ (in particular every λ with $\lambda^{-1} \notin \mathcal{O}_{F_0}$, since $\mathfrak{a} \subseteq \mathcal{O}_{F_0}$), and the shape (Lin). So the propagation statement $\mathrm{P2}_{\mathrm{split}}$ is not closed, and nothing here bears on (F2) or (F3') (§6). Section 7 checks and folds in two earlier working notes, and §8 lists the computations.

2. SETUP

Setup 2.1. We use the notation of [BMB, §19]. F is a quartic CM field with totally real subfield F_0 , real places τ_1, τ_2 and nontrivial automorphism ι of F_0 ; for $x \in F_0$ write $x_t = \tau_t(x)$. A is a member of an $(F, 2, \delta)$ -family: an abelian eightfold with an order of F in $\text{End}(A)$ and a polarisation L with class θ . Over $\mathbb{R}$, $H^1(A, \mathbb{R}) = H_1 \oplus H_2$ with F_0 acting on H_t through τ_t , and $H_t \otimes \mathbb{C} = V_\sigma \oplus V_{\bar{\sigma}}$ for the two embeddings $\sigma, \bar{\sigma}$ of F over τ_t ; each V_σ has dimension 4, with $V_\sigma^{1,0}$ and $V_\sigma^{0,1}$ of dimension 2. Since F_0 is fixed by the Rosati involution, $\theta = \theta_1 + \theta_2$ with $\theta_t \in \Lambda^2 H_t$ symplectic on H_t , and θ_t pairs V_σ with $V_{\bar{\sigma}}$. For $g \in F_0[[T]]$ put $\text{Tr}(g(\Theta)) = g^{\tau_1}(\theta_1) + g^{\tau_2}(\theta_2)$, with g^{τ_t} obtained by applying τ_t to the coefficients; so $\eta_s = \text{Tr}(s\Theta) = s_1\theta_1 + s_2\theta_2$, and $\text{Tr}(g(\Theta))$ is a rational class when g has coefficients in F_0 . The line $\Lambda^4 V_\sigma$ is spanned by α_σ , and $\alpha_\Sigma = \prod_{\sigma \in \Sigma} \alpha_\sigma$. A nonzero rational Weil class is $\omega = \sum_\sigma u_\sigma \alpha_\sigma$ with every $u_\sigma \neq 0$; its place parts are $\omega^{(t)} = u_\sigma \alpha_\sigma + u_{\bar{\sigma}} \alpha_{\bar{\sigma}}$.

For $\gamma \in H^*(A, \mathbb{C})$, $r(\gamma)$ is the rank of $HT^2(A) \rightarrow H^*(A, \mathbb{C})$, $\xi \mapsto \xi \lrcorner \gamma$, where $HT^2(A) = \Lambda^2 H^{0,1} \oplus H^{0,1} \otimes T \oplus \Lambda^2 T$ and $T = H^0(T_A)$. For a perfect complex E with $\text{ch}(E) = N\omega + P$, $N \neq 0$ and P a rational polynomial in the η_s , [BMB, Theorem 19.21(v)] gives

$$(1) \quad \dim \text{Ext}^2(E, E) \geq r(\text{ch } E) \geq 80,$$

and equality in the first inequality, which we call *the criterion*, implies $\mathbf{W}(F, 2, \delta)$: the Weil classes are algebraic on every abelian variety of $(F, 2, \delta)$ -Weil type. The rank is computed in closed form in [BMB, Theorem 19.22]; the pure shape $c + N\omega$, of rank 80, is excluded by [BMB, Theorem 19.12], so the least rank of a complex meeting the criterion is 88, and at 88 the two shapes of §1 are all that remain [BMB, Proposition 19.24, Corollary 19.26].

The Weil tori are the complex tori S_F of [BMB, Setup 19.9]: the complex structures on $H^1(A, \mathbb{R})$ commuting with F with each $V_\sigma^{1,0}$ of dimension 2. The classes α_Σ are of Hodge type on every member of S_F , and [BMB, Theorem 19.12] states: *if $\text{ch}(E)$ stays of Hodge type on every member of S_F and $\text{ch}_2(E) \neq 0$, then $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$.*

3. THE PLACE-WISE MUKAI GROUP

Definition 3.1. For $t = 1, 2$ let L_t be the product with θ_t on $H^*(A, \mathbb{R}) = \Lambda^* H_1 \otimes \Lambda^* H_2$, let $\pi_t \in \Lambda^2 H_t^\vee$ be the bivector inverse to θ_t , let $\Lambda_t = -\iota_{\pi_t}$ be minus the contraction with it, normalised so that $\Lambda_t \theta_t = 4$, and let H_t act on $\Lambda^k H_t$ by $k - 4$.

Lemma 3.2. *Each (L_t, H_t, Λ_t) is an $\mathfrak{sl}_2$ -triple, and the operators of the two places commute. On the span of the powers of θ_t the assignment $\theta_t^k/k! \mapsto \binom{4}{k} x^{4-k} y^k$ identifies the action with that on binary quartics, $L_t = y \partial_x$ and $\Lambda_t = x \partial_y$; in particular $e^{\lambda \theta_t} \mapsto (x + \lambda y)^4$, and 1 and $\theta_t^4/24$ go to x^4 and y^4 .*

Proof. The first statement is the Lefschetz $\mathfrak{sl}_2$ of the symplectic space (H_t, θ_t) of dimension 8, acting on the factor $\Lambda^* H_t$; operators on different tensor factors commute, being even. The span of the powers of θ_t is the irreducible representation generated by 1, of highest weight 4, and $\Lambda_t \theta_t^k/k! = (5-k) \theta_t^{k-1}/(k-1)!$ matches $x \partial_y \binom{4}{k} x^{4-k} y^k = (5-k) \binom{4}{k-1} x^{5-k} y^{k-1}$. Items (A) of §8 check the relations. $\square$

Lemma 3.3. *$\Lambda_t \alpha_\sigma = 0$ for every σ and t , and $L_t \alpha_\sigma = 0$ for σ over τ_t . So $\omega^{(t)}$ is invariant under the $\mathfrak{sl}_2$ of its own place and is a lowest weight vector, of weight -4 , for the other.*

Proof. For σ over τ_t , V_σ is a Lagrangian subspace of $(H_t \otimes \mathbb{C}, \theta_t)$, since θ_t pairs it with $V_{\bar{\sigma}}$ and it has half the dimension. Every term of θ_t has a factor in V_σ , which repeats one of α_σ , so $\theta_t \alpha_\sigma = 0$; and π_t contracts a factor from V_σ with one from $V_{\bar{\sigma}}$, of which α_σ has none. At the other place α_σ has degree 0, hence weight -4 , and $\Lambda_{t'}$ lowers the degree. Item (B) checks this. $\square$

Definition 3.4. Let $\varphi_L: A \rightarrow \hat{A}$ be the polarisation. Put

$$\mathcal{O}' = \{b \in F_0 : \varphi_L \circ b \in \text{Hom}(A, \hat{A})\}, \quad \mathfrak{a} = \{\beta \in F_0 : \beta \circ \varphi_L^{-1} \in \text{Hom}(\hat{A}, A)\}.$$

Both are subgroups of F_0 . If e is the exponent of $\ker \varphi_L$ and $R = F_0 \cap \text{End}(A)$, then $R \subseteq \mathcal{O}' \subseteq e^{-1}R$ and $eR \subseteq \mathfrak{a} \subseteq R \subseteq \mathcal{O}_{F_0}$; so both are lattices of rank 2, and both are R -modules. If L is principal, $\mathfrak{a} = \mathcal{O}' = R$.

The inclusions: $e = \psi \circ \varphi_L$ for a homomorphism ψ , which is $e\varphi_L^{-1}$; $\beta \circ \varphi_L^{-1} \in \text{Hom}(\hat{A}, A)$ gives $\beta = (\beta \circ \varphi_L^{-1}) \circ \varphi_L \in \text{End}(A)$; and $\varphi_L \circ b \in \text{Hom}(A, \hat{A})$ gives $eb = \psi \circ (\varphi_L \circ b) \in \text{End}(A)$. Composing with elements of R preserves both conditions.

Proposition 3.5. (i) For $b \in \mathcal{O}'$ there is a line bundle L_b on A with $c_1(L_b) = \eta_b$, and $T_b = L_b \otimes (-)$ acts on $H^*(A, \mathbb{Q})$ by $\exp(b_1 L_1 + b_2 L_2)$.
(ii) For $\beta \in \mathfrak{a}$ there is a line bundle M_β on $\hat{A}$ with $\varphi_{M_\beta} = \beta \circ \varphi_L^{-1}$, and $S_\beta = \Phi_{\mathcal{P}}^{-1} \circ (M_\beta \otimes -) \circ \Phi_{\mathcal{P}}$, with $\Phi_{\mathcal{P}}$ the Fourier–Mukai transform of the Poincaré bundle, is an autoequivalence of $D^b(A)$ acting by $\exp(\epsilon(\beta_1 \Lambda_1 + \beta_2 \Lambda_2))$ for a sign $\epsilon = \pm 1$ that does not depend on A or β .
Write $G(A)$ for the group generated by the T_b and the S_β .

Proof. On a complex abelian variety the Néron–Severi group is the group of symmetric homomorphisms to the dual [BL04, §2.5]. Elements of F_0 are fixed by the Rosati involution, so $\varphi_L \circ b$ and $\beta \circ \varphi_L^{-1}$ are symmetric; when they are homomorphisms they are φ of line bundles. The class of L_b is the form $\theta(b \cdot, \cdot)$, which is $b_t \theta_t$ on H_t , so $\text{ch}(L_b) = e^{\eta_b}$ and (i) follows, L_1 and L_2 commuting.

For (ii), $\Phi_{\mathcal{P}}$ is an equivalence by Mukai [Muk81], and on cohomology it is $x \mapsto p_*(q^*x \cdot e^\ell)$ with ℓ the Poincaré class, the Todd classes being trivial [BMB, Proposition 6.1]. In a basis e_i of $H^1(A)$ with dual basis f_i of $H^1(\hat{A})$, $\ell = \sum e_i f_i$ and $f_i e^\ell = \partial_{e_i} e^\ell$; since p_* extracts the top coefficient in the e_i and kills derivatives, integration by parts gives $f_i \cdot \Phi(x) = \pm \Phi(\iota_{e_i^\vee} x)$, and for a class $y \in \wedge^2 H^1(\hat{A})$,

$$y \cdot \Phi_{\mathcal{P}}(x) = \Phi_{\mathcal{P}}(\iota_y x), \quad \text{so} \quad \Phi_{\mathcal{P}}^{-1}(e^y \Phi_{\mathcal{P}}(x)) = \exp(\iota_y) x,$$

with y read as a bivector on $H^1(A)$ through ℓ (item (K) checks the identity for abelian varieties of dimension at most three, signs included). The class $c_1(M_\beta)$, read in this way, is the bivector of the alternating form of $\beta \circ \varphi_L^{-1}$, which is $\beta_1 \pi_1 + \beta_2 \pi_2$ up to one universal sign, because φ_L^{-1} has bivector $\pi_1 + \pi_2$ up to that sign and F_0 acts on H_t through π_t . For L principal and $\beta = 1$ this is the dual principal polarisation, and $\iota_\pi \theta = \pm 8 = \pm(4 + 4)$, which is the normalisation of Definition 3.1. $\square$

Proposition 3.6. For g in the group generated by the $\exp(xL_t)$ and $\exp(x\Lambda_t)$, $x \in \mathbb{C}$, $r(g\gamma) = r(\gamma)$ for every γ . Hence for every $\Phi \in G(A)$ and every perfect complex E ,

$$\dim \text{Ext}^2(\Phi E, \Phi E) = \dim \text{Ext}^2(E, E), \quad r(\text{ch } \Phi E) = r(\text{ch } E),$$

and E meets the criterion if and only if $\Phi(E)$ does.

Proof. Let $U \subset \text{End } H^*(A, \mathbb{C})$ be the span of the products with elements of $H^{0,1}$ and of the contractions with elements of $T = T^{1,0}$. The operators $\xi_\perp(\cdot)$, $\xi \in HT^2(A)$, span exactly the products of two elements of U . Now $[L_t, \iota_v] = -(\iota_v \theta_t) \wedge$, and $\iota_v \theta_t \in H^{0,1}$ for $v \in T^{1,0}$ because θ_t is of type $(1,1)$; $[L_t, y \wedge] = 0$; $[\Lambda_t, y \wedge]$ is, up to sign, the contraction with $\pi_t(y) \in T^{1,0}$ for $y \in H^{0,1}$, since π_t is of type $(1,1)$; and $[\Lambda_t, \iota_v] = 0$. So $\text{ad } L_t$ and $\text{ad } \Lambda_t$ preserve U , conjugation by g preserves U and $U \cdot U$, and $\dim(U \cdot U) g\gamma = \dim g^{-1}(U \cdot U) g\gamma = \dim(U \cdot U) \gamma$. An equivalence preserves Ext^2 , and it acts on Chern characters by its cohomological action. Item (E) checks the invariance on the fifteen values of [BMB, Theorem 19.22] and on random words. $\square$

4. EXCLUSION OF THE EXPONENTIAL SHAPES

Lemma 4.1. *For $\lambda, \beta \in \mathbb{C}$ and each t ,*

$$\exp(\beta \Lambda_t) e^{\lambda \theta_t} = \begin{cases} (1 + \lambda \beta)^4 e^{\lambda' \theta_t}, & \frac{1}{\lambda'} = \frac{1}{\lambda} + \beta, \quad 1 + \lambda \beta \neq 0, \\ \lambda^4 \theta_t^4 / 24, & 1 + \lambda \beta = 0. \end{cases}$$

Proof. By Lemma 3.2, $\exp(\beta x \partial_y)$ substitutes $y + \beta x$ for y , and $(x + \lambda(y + \beta x))^4 = ((1 + \lambda \beta)x + \lambda y)^4$. Item (C) checks both cases. $\square$

Theorem 4.2. *Let E be a perfect complex on A and $\beta \in \mathfrak{a}$. If $\exp(\beta_1 \Lambda_1 + \beta_2 \Lambda_2) \text{ch}(E)$ stays of Hodge type on every member of S_F and has nonzero component in H^4 , then $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$.*

Proof. $\mathfrak{a} = -\mathfrak{a}$, so with ϵ as in Proposition 3.5 the autoequivalence $\Phi = S_{\epsilon \beta}$ has $\text{ch}(\Phi E) = \exp(\beta_1 \Lambda_1 + \beta_2 \Lambda_2) \text{ch}(E)$. By [BMB, Theorem 19.12], $\dim \text{Ext}^2(\Phi E, \Phi E) > r(\text{ch } \Phi E)$, and both sides are those of E by Proposition 3.6. $\square$

Theorem 4.3. *Let A be a member of an $(F, 2, \delta)$ -family, ω a nonzero rational Weil class, $N \neq 0$, $c, b \in \mathbb{Q}$, $a \in F_0^\times$ and $\lambda \in F_0^\times$, and let E be a perfect complex on A with*

$$\text{ch}(E) = N\omega + c + \text{Tr}(a(e^{\lambda \Theta} - 1)) + b(e^{\eta \lambda} - 1).$$

If $\lambda^{-1} \in \mathfrak{a}$, equivalently if $\varphi_{\eta \lambda}^{-1} = \lambda^{-1} \circ \varphi_L^{-1}$ is a homomorphism $\hat{A} \rightarrow A$, then E does not meet the criterion: $\dim \text{Ext}^2(E, E) > r(\text{ch } E)$. Here $r(\text{ch } E) = 88$ when $b = 0 \neq a$, and $r(\text{ch } E) = 104$ when $b \neq 0$.

Proof. Take $\beta = -\lambda^{-1} \in \mathfrak{a}$. By Lemma 3.3 the operators Λ_t kill ω and 1; by Lemma 4.1 at both places, the operators of the two places acting on separate tensor factors,

$$\exp(\beta_1 \Lambda_1 + \beta_2 \Lambda_2) \text{ch}(E) = N\omega + (c - \text{Tr } a - b) + \frac{1}{24} \text{Tr}(a \lambda^4 \Theta^4) + \frac{b}{576} \text{Nm}(\lambda)^4 \theta_1^4 \theta_2^4.$$

As in the proof of [BMB, Proposition 19.24], θ_t^4 is a nonzero multiple of $\alpha_\sigma \alpha_{\bar{\sigma}}$ for the embeddings over τ_t , so this class lies in the span of the α_Σ , is rational, stays of Hodge type on S_F , and has component $N\omega \neq 0$ in H^4 . Theorem 4.2 applies. For the ranks, in the notation of [BMB, Theorem 19.22] the Hankel matrices of $(a_t \lambda_t^k)$ have rank $\rho_t = 1$, so $R_t = 8$, and $\mu = 0$ exactly when $b = 0$; the formula gives $64 + 4 + 4 + 8 + 8 = 88$ and $88 + 16 = 104$. Items (D) and (F) check the transform and the ranks. $\square$

Corollary 4.4. *Let $\beta \in \mathfrak{a}$ with $1 + \lambda \beta \neq 0$. Then $S_{\pm \beta}$ carries a character of shape (Exp) with parameters (c, a, λ) to one with*

$$a' = a(1 + \lambda \beta)^4, \quad \frac{1}{\lambda'} = \frac{1}{\lambda} + \beta, \quad c' = c + \text{Tr}(a' - a).$$

So the orbits of the shapes (Exp) under the S_β are indexed by λ^{-1} in $F_0/\mathfrak{a}$, an infinite group, and Theorem 4.3 removes exactly the class of 0: a transform S_β puts (Exp) in the span of the α_Σ only when $1 + \lambda \beta = 0$.

Proof. Lemma 4.1. For $1 + \lambda \beta \neq 0$ the transform still has a nonzero component $a' \lambda' \theta$ in the span of θ_1, θ_2 , which is not in the span of the α_Σ . $\square$

Corollary 4.5. *If L is principal and $\mathcal{O}_{F_0} \subseteq \text{End}(A)$, then $\mathfrak{a} = \mathcal{O}_{F_0}$, and Theorem 4.3 excludes (Exp) exactly for $\lambda^{-1} \in \mathcal{O}_{F_0}$: for instance $\lambda = 1$, that is $\text{ch}(E) = N\omega + c + \text{Tr}(a(e^\Theta - 1))$, and $\lambda = 1/m$, $m \in \mathbb{Z} \setminus \{0\}$; not $\lambda = 2$. At any member, $\mathfrak{a} \subseteq \mathcal{O}_{F_0}$, so no λ with $\lambda^{-1} \notin \mathcal{O}_{F_0}$ is excluded by this argument.*

Remark 4.6 (Where the argument stops). The transforms in $G(A)$ act through the group M of matrices $\begin{pmatrix} u & b \\ c & v \end{pmatrix} \in \mathrm{SL}_2(F_0)$ with $u, v \in R$, $b \in \mathcal{O}'$ and $c \in \mathfrak{a}$, embedded by (τ_1, τ_2) ; it is a group because $\mathcal{O}'$ and $\mathfrak{a}$ are R -modules and $\mathcal{O}'\mathfrak{a} \subseteq R$. At each place the Weil part of the other place, a lowest weight vector, pins the line of x , so a transform that puts (Exp) into the span of the α_Σ must send the pair of lines $\{x, x + \lambda y\}$ to $\{x, y\}$ at both places. If it fixes the line of x , it is lower triangular with a unit on the diagonal and an entry of $\mathfrak{a}$ below it, and it sends $x + \lambda y$ to the line of y exactly when $\lambda^{-1} \in \mathfrak{a}$: that is Theorem 4.3. If it exchanges the two lines, it moves $\omega^{(t)} \otimes x^4$ to $\omega^{(t)}\theta_t^4/24$, so the transformed character has no component in H^4 and a nonzero one in H^{12} ; [BMB, Theorem 19.12] does not apply, since the vanishing on a very general Weil torus [BMB, Proposition 19.11] covers ch_k for $1 \leq k \leq 3$ only. When L is principal, the Weyl element lies in M , such an exchange is the Weyl element times a transform of the first kind, and it reaches no further λ . So Theorem 4.3 is all this group gives on (Exp), and $\lambda^{-1} \notin \mathfrak{a}$ needs another idea.

5. THE LINEAR SHAPE

Throughout, $\gamma = N\omega + c + \eta_s$ with $N \neq 0$, $c \in \mathbb{Q}$ and $s \in F_0^\times$, so $s_1 s_2 \neq 0$.

Proposition 5.1. *No element of $\mathrm{SL}_2(\mathbb{C}) \times \mathrm{SL}_2(\mathbb{C})$, acting through the two $\mathfrak{sl}_2$ of Lemma 3.2, carries γ into the span of the α_Σ . In particular no $\Phi \in G(A)$ does, and Theorem 4.2 does not apply to (Lin).*

Proof. The group preserves the isotypic components of $H^*(A)$ for $\mathfrak{sl}_2 \times \mathfrak{sl}_2$. The component of $\mathrm{Sym}^4 \otimes \mathrm{Sym}^4$ is the span of the monomials $\theta_1^i \theta_2^j$, because the only class of weight $(-4, -4)$ is the constant; its intersection with the span of the α_Σ is spanned by x^4 and y^4 at each place, that is by 1, θ_1^4 , θ_2^4 and $\theta_1^4 \theta_2^4$, while the other α_Σ lie in other components (Lemma 3.3). The component of γ there is

$$p = c x_1^4 x_2^4 + 4s_1 x_1^3 y_1 x_2^4 + 4s_2 x_1^4 x_2^3 y_2.$$

Read p as a map from the dual of the quartics of the second place to those of the first; its image contains $x_1^3 y_1$ because $s_1 s_2 \neq 0$. If $(g_1, g_2)p$ lay in $\langle x^4, y^4 \rangle^{\otimes 2}$, the image would lie in $\langle m^4, n^4 \rangle$ with $m = g_1^{-1}x$, $n = g_2^{-1}y$ independent. But a nonzero $um^4 + vn^4$ is a fourth power when $uv = 0$ and has four distinct roots otherwise, its discriminant being $256u^3 v^3 \det(m, n)^{12}$, while $x^3 y$ has a triple root. Item (H) checks the discriminant and the image. $\square$

The exponential shape corresponds to the fourth power $(x + \lambda y)^4$, which the group can move to y^4 ; the linear shape corresponds to its derivative at $\lambda = 0$, which it cannot.

Proposition 5.2. *For $\beta \in \mathfrak{a}$, $S_{\pm\beta}(E)$ has character $N\omega + (c + 4\mathrm{Tr}(s\beta)) + \eta_s$. So the rank of a complex of shape (Lin) meeting the criterion can be moved by any element of $4\mathrm{Tr}(s\mathfrak{a}) \subseteq \mathbb{Z}$ without leaving the shape or the criterion; rank 0 is reached exactly when $c \in 4\mathrm{Tr}(s\mathfrak{a})$. The β with $\mathrm{Tr}(s\beta) = 0$ form a subgroup of rank one of $\mathfrak{a}$, which fixes the character.*

Proof. $\exp(\beta\Lambda_t)\theta_t = \theta_t + 4\beta$ and Λ_t kills 1 and ω ; Proposition 3.6. Item (G). $\square$

Proposition 5.3. *Let L be principal, so that $1 \in \mathcal{O}' \cap \mathfrak{a}$, and let $W = T_1 S_{-\epsilon} T_1$, which acts by $w = \exp(L)\exp(-\Lambda)\exp(L)$ at both places. For every $\beta \in \mathfrak{a}$ and every E with $\mathrm{ch}(E) = \gamma$, the object $WS_\beta(E)$ is not isomorphic to a shift of a coherent sheaf.*

Proof. At each place w sends $x \mapsto y$ and $y \mapsto -x$; so $w(1)$ is the class $\theta_1^4 \theta_2^4 / 576$ of a point, $w(\theta_t)$ has degree 14, and $w(\omega) = (\omega^{(1)}\theta_2^4 + \theta_1^4\omega^{(2)})/24$. By Proposition 5.2 the character of $WS_\beta(E)$ therefore vanishes below degree 12 and has

$$\mathrm{ch}_6 = \frac{N}{24}(\omega^{(1)}\theta_2^4 + \theta_1^4\omega^{(2)}) \neq 0, \quad \int_A \mathrm{ch}_6 \cdot \theta_i \theta_j = 0 \quad \text{for all } i, j,$$

the integrals vanishing because $\theta_t \omega^{(t)} = 0$ (Lemma 3.3) and $\theta_t^5 = 0$. If $WS_\beta(E) \cong \mathcal{F}[k]$ for a sheaf $\mathcal{F}$, then $\text{ch}(\mathcal{F})$ vanishes below degree 12; the lowest nonzero Chern character of a sheaf is the class of the cycle of its support in that codimension [Ful98, §15.1, Ex. 15.1.5 and Cor. 18.3.2], which is effective, so $\mathcal{F}$ is supported in dimension at most two and $\text{ch}_6(\mathcal{F})$ is a nonzero effective cycle, whose integral against θ^2 is positive. This contradicts the vanishing. Item (I) checks w and the integrals. $\square$

So a complex of shape (Lin) meeting the criterion cannot be produced as the inverse transform, by WS_β , of a sheaf on a surface: the object must be built as a genuine complex on that side as well.

Proposition 5.4. *Suppose $A = B \times B'$ with B, B' abelian fourfolds of $(F, 1)$ -Weil type and the product structure. Then the component of $\text{ch}_2(E) = N\omega$ in $H^2(B) \otimes H^2(B')$ has tensor rank 4. Hence if $[E] = \sum_{i=1}^k \pm [E_i \boxtimes E'_i]$ in $K_0(A)_{\mathbb{Q}}$, then $k \geq 4$; in particular E is not an exterior product, nor a cone of two.*

Proof. $\bigwedge^4 V_\sigma = \bigwedge^2 V_\sigma(B) \otimes \bigwedge^2 V_\sigma(B')$, so each α_σ is a pure tensor, and the four lie in different eigenspaces of F on both sides; the θ_t have no component there. The component of $\text{ch}(E_i \boxtimes E'_i)$ is $c_1(E_i) \otimes c_1(E'_i)$. Item (J). $\square$

Proposition 5.5. *Let $E = \bigoplus_i E_i$ meet the criterion, with each $\text{ch}(E_i)$ of the form $N_i\omega + P_i$, P_i a rational polynomial in the η_s . Then each E_i meets its own criterion, $\text{Ext}^2(E_i, E_j) = 0$ for $i \neq j$, and $r(\text{ch } E) = \sum_i r(\text{ch } E_i)$. If $r(\text{ch } E) < 108$, exactly one summand has nonzero Chern character.*

Proof. $\dim \text{Ext}^2(E, E) \geq \sum_i \dim \text{Ext}^2(E_i, E_i) \geq \sum_i r(\text{ch } E_i) \geq r(\text{ch } E)$, the last because a rank is subadditive; equality throughout gives the first three statements. By [BMB, Theorem 19.21(iii)], $r(N_i\omega + P_i) \geq 80$ when $N_i \neq 0$. When $N_i = 0 \neq P_i$, $r(P_i) \geq 28$: the torus scaling the classes of type $(1, 0)$ at one place multiplies θ_t alone and normalises U , so it keeps r , and degenerating P_i to a vertex of its Newton polygon, r being lower semicontinuous, gives $r(P_i) \geq \min_{i,j} r(\theta_1^i \theta_2^j) = 28$ (item (L)). Since $\sum N_i = N \neq 0$, a second summand with nonzero character would give $r \geq 80 + 28$. $\square$

6. WHAT REMAINS

- (a) $P2_{\text{split}}$ at a quartic field. At rank 88: the shape (Exp) with $\lambda^{-1} \notin \mathfrak{a}$, which includes every λ with $\lambda^{-1} \notin \mathcal{O}_{F_0}$, and the shape (Lin), subject to Propositions 5.1–5.5. Above 88, only the mixed exponential of rank 104 with $\lambda^{-1} \in \mathfrak{a}$ is removed here. Within $G(A)$ the only further gain would come from the vanishing of ch_6 on a very general Weil torus (Remark 4.6) together with a transform exchanging the two lines, which exists only for non-principal L . What (Lin) needs is an object: a complex, not a shifted sheaf after WS_β , not a single summand of lower rank, and on a product member not fewer than four exterior products in K -theory.
- (b) (F2). The mechanism of §3 needs a totally real field in $\text{End}(A)$ to split the Lefschetz $\mathfrak{sl}_2$ into places. A Mumford fourfold has $\text{End} = \mathbb{Z}$, its exceptional classes are rigid [BMB, Theorem 21.21], and every object built from line bundles by cones, tensor products, duals, pull-backs, push-forwards and Fourier–Mukai functors with such kernels has a Chern character invariant under the Lefschetz group, which no exceptional class is [BMB, Theorem 21.36]. So no transform of the kind used here reaches (F2), and we have no construction for it.
- (c) (F3'). Nothing here bears on it.

7. EARLIER WORKING NOTES

7.1. The smooth-support note. A working note of the same date, *Record of an attempt on the smooth-support case at $n = 4$* , concerns an imaginary quadratic field $K = \mathbb{Q}(\sqrt{-d})$ and a secant object $\mathcal{I}_S(3\Theta)$ with smooth support on a principally polarised abelian fourfold (X, Θ) , the shape isolated in [BMB, Corollary 18.57, Remark 18.58] for Markman’s construction [Mar25a, Mar25b]. That route is not needed by any minimal sufficient set of the closure graph, since (P2) for the imaginary quadratic fields follows from $P_{2,\text{split}}$ by [BMB, Proposition 20.13] and [BMB, Theorem 20.58]; it is recorded because what it proves is correct. We checked its numerical claims in exact arithmetic (the second program of §8, twenty-five checks) and read its proofs; the one claim not checked, that the degree of the Gauss map divides $\gcd([S]^2, e(S))$, is not used here. Write $u_\Theta = \sum_k (-d)^k \Theta^{2k} / (2k)!$ and $v_\Theta = \sum_k (-d)^k \Theta^{2k+1} / (2k+1)!$, let $\Lambda_\Theta \subset \mathbb{Q}[\Theta] / (\Theta^5)$ be the group generated by the $e^{j\Theta}$, and for $\xi = \sum_i a_i \Theta^i / i!$ define the binomial moments $m_k(\xi)$ by $a_i = \sum_k S(i, k) k! m_k$.

Theorem 7.1 (From the smooth-support note). $\xi \in \Lambda_\Theta$ if and only if $m_0, \dots, m_4 \in \mathbb{Z}$, and

$$m(a u_\Theta + b v_\Theta) = \left(a, b, -\frac{ad+b}{2}, \frac{3ad+2b-bd}{6}, \frac{ad^2-11ad+6bd-6b}{24} \right).$$

In particular $u_\Theta + 3v_\Theta \in \Lambda_\Theta$ if and only if $d \equiv 15$ or $23 \pmod{24}$.

The proof is that $e^{j\Theta} \mapsto (1+y)^j$ identifies Λ_Θ with $\mathbb{Z}[y]/(y^5)$ and the moments with the coefficients. It explains [BMB, Table 4]: a split Hilbert–Burch resolution has its character in Λ_Θ , so only $d \equiv 15, 23 \pmod{24}$ can occur, at any rank and with any twists, which is what the search found below 24. For the four discriminants $d \in \{1, 3, 5, 7\}$ of a smooth support [BMB, Corollary 18.41] it excludes every resolution $0 \rightarrow E_1 \rightarrow E_0 \rightarrow \mathcal{I}_S \rightarrow 0$ with $\text{ch } E_0, \text{ch } E_1 \in \Lambda_\Theta$, for instance by bundles built from line bundles algebraically equivalent to multiples of Θ and from the kernel bundles $M_{m\Theta}$ by sums, tensor products, duals, exterior and symmetric powers, extensions, kernels of surjections and cokernels of injections. These need not be projectively flat, so for the smooth shape this goes beyond [BMB, Theorems 18.52 and 18.55]. One phrase of the original needs correcting: it lists kernels and cokernels of arbitrary maps, but every bundle is the image of a map of constant rank between sums of line bundles, so those kernels and cokernels can have any class; only the exact cases above stay in Λ_Θ , and they are the ones the corollary uses.

The note also computes $m(\mathcal{O}_S) = (0, 0, N, -3N, N(47 - 2N)/12 + 3N)$, whose last entry is not an integer for $N \in \{5, 6, 7, 8\}$, and reproduces [BMB, Table 3] from Grothendieck–Riemann–Roch: $K_S^2 = 12N(36 - N)$, $e(S) = 12N(36 - 3N)$, $\chi(\mathcal{O}_S) = 4N(18 - N)$.

Lemma 7.2 (From the smooth-support note). *Let X be simple, for instance with $\text{NS}(X)_\mathbb{Q} = \mathbb{Q}\Theta$ as in [BMB, Setup 18.47], and $S \subset X$ a smooth surface with $e(S) \neq [S]^2$. Then $H^2(\mathcal{O}_X) \rightarrow H^2(\mathcal{O}_S)$ is injective. For a smooth secant support $[S]^2 = 24N^2 \neq 12N(36 - 3N) = e(S)$ for every integer $N \geq 1$, so the injectivity required in [BMB, Corollary 18.57(ii)] holds automatically.*

The proof: by conjugation the kernel is that of the restriction of invariant 2-forms; a form of rank two vanishing on S gives, by Castelnuovo–de Franchis, a fibration whose fibres generate an abelian subvariety of dimension at most two, against simplicity; a form of rank four vanishing on S makes S Lagrangian, so $N_{S/X} \cong \Omega_S^1$ and $[S]^2 = c_2(N_{S/X}) = e(S)$.

The note further shows that $\delta = K_S - 4\lambda$, $\lambda = \Theta|_S$, is a nonzero algebraic class with $\iota_*\delta = 0$ and $\delta^2 = -12N(N - 4)$, and for general b that $K_S - \frac{4b}{3}\lambda$ has square $\frac{4N}{3}(4b^2 - 9N)$; and, through Bloch’s semiregularity map [Blo72], that S extends to first order along every polarised deformation of X if and only if that map has rank 6, with $h^1(N_{S/X}) \geq 88, 296, 568, 904$ for $N = 5, 6, 7, 8$. All of this is correct; in the lower bound $h^0(N_{S/X}) \geq 4$ the reason is that a translation field tangent to S would make S invariant under a positive-dimensional abelian subvariety, which simplicity

excludes. The note leaves the existence of S and the first-order condition open, as does the manuscript.

Its last section, on the Kuga–Satake class of the Mumford square, proves that for real multiplication of odd degree every Hodge similarity is a rational scalar; as that note says, this is already in the manuscript. Of the works it cites there, those of Morrison [Mor84], Buskin [Bus19] and Varesco [Var23] are in the manuscript’s bibliography; the others are not repeated here.

7.2. The status note. A second note searched, by simulated annealing, unions of coordinate abelian surfaces in $E_1 \times \cdots \times E_4$ for configurations with the numerical invariants of a secant object at $b = 1$. It reports exact matches at $d = 1$ and none at $d = 2, 3$. Its own caveats stand: $E_1 \times \cdots \times E_4$ is not a member of a Weil family with the required action, and the search does not touch the second condition of [BMB, Theorem 18.29]. It is a data point about the imaginary quadratic route, which, as above, the closure graph does not need.

8. COMPUTATIONS

Two programs in exact rational arithmetic accompany this note. They are kept with its source in the repository <https://github.com/creelie/H8>, in the directory `notes/fourier_mukai`. The first reuses the exterior-algebra model of [BMB, item (LXV)] and checks: (A) the two $\mathfrak{sl}_2$ -triples and their commutation; (B) Lemma 3.3; (C) Lemma 4.1; (D) the transform in the proof of Theorem 4.3 on five characters, and that it lands in the span of the α_Σ ; (E) Proposition 3.6 on the fifteen values of [BMB, Theorem 19.22] and on random words; (F) the ranks 88 and 104; (G) Proposition 5.2; (H) the discriminant and the image in Proposition 5.1; (I) the Weyl element and the integrals of Proposition 5.3; (J) Proposition 5.4; (K) the Fourier identity of Proposition 3.5; (L) the least rank 28 over the monomials. It prints `30 checks passed, 0 failed`. The second checks the numerical statements of §7.1, among them the moment formula, the congruence for every $d < 2400$, the four rows of [BMB, Table 3], the first candidates of [BMB, Table 4], and the bounds on $h^1(N_{S/X})$; it prints `25 checks passed, 0 failed`.

Data availability. The two programs and the sources of this note are in the repository above; the manuscript’s code is archived on Zenodo under [10.5281/zenodo.23038895](https://zenodo.org/record/23038895) (release v2.0.0; all versions [10.5281/zenodo.22950275](https://zenodo.org/record/22950275)). The programs were written and run with the assistance of Claude Code (Anthropic).

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